

# Clinic Management System

## System Documentation

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Version 1.0

2 April 2026

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*Multi-Branch SaaS Clinic Management Solution*

*Laravel 12 + Star Admin Bootstrap 4 + Alpine.js + MySQL*

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# 1. System Overview

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The Clinic Management System (CMS) is a multi-branch SaaS web application designed for Malaysian private clinics. It manages the full clinic workflow - from patient registration, doctor scheduling, and appointments to billing, pharmacy, lab reports, insurance claims, and patient self-service portal.

## Key Highlights

- Multi-Branch: Centralized system with branch-based data filtering and branch switcher
- Role-Based Access: 4 user roles - Admin, Doctor, Staff, Receptionist
- Complete Clinical Workflow: Appointments ->Prescriptions ->Pharmacy ->Lab ->Billing ->Insurance Claims
- Patient Portal: Separate login for patients to view their records
- Audit Trail: Every create/update/delete action logged with full diff
- In-App Notifications: Real-time alerts for appointments, payments, low stock, etc.
- Reports with Export: 5 report modules with CSV export

## 2. Tech Stack

| Component      | Technology                                      |
|----------------|-------------------------------------------------|
| Framework      | Laravel 12 (PHP 8.4)                            |
| Frontend       | Blade Templates + Star Admin Bootstrap 4        |
| JavaScript     | Alpine.js (dynamic forms)                       |
| Database       | MySQL / MariaDB (port 3307)                     |
| Authentication | Laravel Breeze (staff) + Session-based (portal) |
| File Storage   | Laravel Storage (public disk)                   |
| Icons          | Material Design Icons (MDI)                     |

## Directory Structure

```
clinic-app/  
|-- app/Http/Controllers/      # 21 controllers  
|-- app/Http/Middleware/      # AdminMiddleware, RoleMiddleware, PortalAuth  
|-- app/Models/               # 27 Eloquent models  
|-- app/Traits/               # Auditable, NotifiesUsers  
|-- database/migrations/      # 33 migration files  
|-- database/seeders/         # Full test data seeder  
|-- resources/views/          # 90+ Blade views  
|-- routes/web.php             # ~150 routes  
|-- public/star-admin/        # Template assets (CSS/JS)
```

## 3. Installation & Setup

### Requirements

- PHP 8.4+
- Composer
- MySQL / MariaDB
- Node.js (not required - no Vite/Tailwind build)

### Installation Steps

```
# 1. Install dependencies
cd clinic-app && composer install

# 2. Environment setup
cp .env.example .env
php artisan key:generate

# 3. Configure .env
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3307
DB_DATABASE=clinic_management
DB_USERNAME=root
DB_PASSWORD=

# 4. Migrate and seed
php artisan migrate --seed

# 5. Storage link
php artisan storage:link

# 6. Serve
php artisan serve
```

### Default Login Accounts

| Role   | Email                   | Password |
|--------|-------------------------|----------|
| Admin  | admin@clinic.com        | password |
| Doctor | dr.ahmad@clinic.com     | password |
| Doctor | dr.siti@clinic.com      | password |
| Doctor | dr.lee@clinic.com       | password |
| Staff  | nurse.aminah@clinic.com | password |

#### Patient Portal Login:

| Patient | IC Number | Password |
|---------|-----------|----------|
|---------|-----------|----------|

|                  |                |            |
|------------------|----------------|------------|
| Fatimah Abdullah | 850215-14-5678 | patient123 |
|------------------|----------------|------------|

## 4. User Roles & Access Control

### Roles

| Role         | Description           | Access Level                    |
|--------------|-----------------------|---------------------------------|
| Admin        | Full system access    | All modules + System settings   |
| Doctor       | Clinical access       | Clinical modules (own schedule) |
| Staff        | Day-to-day operations | Operational modules             |
| Receptionist | Front desk            | Patient-facing modules          |

### Access Restrictions

- User Management - Admin only
- Audit Logs - Admin only
- Settings - All authenticated users
- All other modules - All authenticated roles
- Patient Portal - Separate authentication (IC + password)

### Middleware

| Middleware | Purpose                                     |
|------------|---------------------------------------------|
| auth       | Standard Laravel authentication             |
| admin      | Restricts to admin role only                |
| role:x,y   | Restricts to specified roles (variadic)     |
| PortalAuth | Patient portal session-based authentication |

## 5. System Modules

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### 5.1 Branch Management

Manages multiple clinic locations. Each branch has its own data scope.

- Fields: Name, Code (unique), Address, Phone, Email, Opening/Closing Time, Status
  - Branch Switcher: Dropdown in navbar to filter data by branch
  - Session-based: `current_branch_id` stored in session
- 

### 5.2 Patient Management

Central patient registry with Malaysian-specific fields.

- Fields: Name, IC Number, Gender, DOB, Phone, Email, Address, Emergency Contact, Allergies, Medical History, Blood Type
  - Auto-Generated Patient ID: Format {branch\_code}-P{number} (e.g., KU-SA-P00001)
  - Portal Access: Admin can enable patient portal login
  - Patient Profile: View appointments, invoices, prescriptions, lab reports, insurance coverage
- 

### 5.3 Doctor & Schedule Management

Manage permanent doctors and their weekly schedules.

- Doctor Fields: User account, Branch, Specialization, Qualification, MMC Number, APC Number, Consultation Fee
  - Schedule: Weekly grid - set available days, start/end time, slot duration (e.g., 30 min)
  - Auto-creates User: Creating a doctor auto-creates a user account with doctor role
- 

### 5.4 Appointment Management

Book and manage patient appointments.

- Fields: Patient, Doctor, Date, Time Slot, Reason, Notes
  - Status Flow: pending ->confirmed ->in\_progress ->completed
  - Can also mark as: cancelled or no\_show
  - Linked To: Invoice, Prescription, Lab Report (one-to-one from appointment)
  - Filters: By date, doctor, status
-



## 5.5 Services & Billing

### Services

- Fields: Name, Description, Price, Category, Status
- Categories: Consultation, Lab, Imaging, Diagnostic, Procedure, Treatment, Document

### Invoices

- Auto-Generated Invoice Number: Format INV-{branch\_code}-{number}
- Dynamic Line Items: Add services with quantity and price (Alpine.js)
- Calculations: Subtotal, Tax, Discount, Total
- Payment Types: Cash or Insurance Panel
- Status: draft ->issued ->partial ->paid
- Print View: Printable invoice layout

### Payments

- Methods: Cash, Card, Bank Transfer, E-Wallet
- Partial Payments: Multiple payments per invoice
- Auto-Status: Invoice status auto-updates based on payment total

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## 5.6 Insurance Panel & Claims

Full insurance workflow - panel management, patient coverage, GL support, claims.

### Insurance Panels

- Types: Corporate, Insurance, TPA, Government
- Fields: Company Name, Contact, Credit Terms, Consultation Limit, Annual Limit, Covered Services, Exclusions, GL Required

### Patient Insurance Coverage

- Fields: Member ID, Policy Number, Employer, Department, Effective/Expiry Date, Remaining Limit, Status

### Insurance Claims

- Auto-Generated Claim Number: Format CLM-{branch\_code}-{number}
  - GL (Guarantee Letter): If panel requires GL, claim starts with gl\_status: pending
  - Status Flow: draft ->submitted ->approved / partial / rejected ->paid
  - Fields: Claim Amount, Approved Amount, Patient Co-pay, Dates, Rejection Reason
-

## 5.7 Pharmacy & Prescriptions

### Medicines

- Fields: Name, Generic Name, SKU, Unit, Cost/Selling Price, Reorder Level, Current Stock, Expiry Date
- Alerts: Low stock warning (stock <= reorder level), Expiry tracking
- Stock Adjustment: Purchase, Adjustment, Return, Expired - all tracked with stock movements

### Prescriptions

- Fields: Patient, Doctor, Appointment, Status, Notes
  - Items: Medicine, Dosage, Frequency, Duration, Quantity, Instructions
  - Dispense Action: Marks as dispensed + auto-deducts stock from pharmacy
  - Status: draft ->dispensed
- 

## 5.8 Lab Tests & Reports

### Lab Tests

- Fields: Name, Description, Category, Normal Range, Unit, Price
- Categories: Blood, Urine, Imaging, etc.

### Lab Reports

- Auto-Generated Report Number: Format LAB-{branch\_code}-{number}
  - Items: Each report contains multiple test results
  - Result Entry: Value, Normal/Abnormal flag, Notes
  - Status: pending ->in\_progress ->completed
  - Print View: Printable lab report with results table
- 

## 5.9 Locum Doctor Management

Manage temporary/contract doctors and their sessions.

- Locum Doctors (Global): Name, IC, MMC/APC, Specialization, Hourly Rate, Session Rate, Bank Details
  - Locum Sessions (Branch-scoped): Date, Start/End Time, Status, Total Pay, Is Paid
  - Pay Calculation: session\_rate if set, otherwise hourly\_rate x hours worked
  - Status: scheduled ->in\_progress ->completed
- 

## 5.10 Appointment Reminders

- Channels: WhatsApp (primary), SMS, Email
- Bulk Create: Auto-generate reminders for all appointments on a target date

- Status: pending ->sent / failed
  - Note: WhatsApp API currently in simulated mode
- 

## 5.11 Settings

| Setting     | Description                  |
|-------------|------------------------------|
| clinic_name | Displayed in navbar & footer |
| clinic_logo | Logo image in navbar         |

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## 5.12 User Management (Admin Only)

- CRUD: Create, edit, deactivate users
  - Assign Roles: Admin, Doctor, Staff, Receptionist
  - Assign Branch: Link user to a branch
  - Activate/Deactivate: Inactive users cannot log in
  - Public registration is disabled - only admin can create users
- 

## 5.13 Profile Management

- Editable: Name, Phone, Profile Photo
- Read-Only: Email, Role, Branch, Member Since
- Profile Photo: Upload JPG/PNG/WebP (max 2MB), preview, remove option
- Change Password: Current password required

## 6. Database Schema

### Entity Relationship Summary

```

Branch (1) ---- (M) Patient, Doctor, Appointment, Invoice, Service, Medicine
User   (1) ---- (1) Doctor
Doctor (1) ---- (M) DoctorSchedule, Appointment
Patient(1) ---- (M) Appointment, Invoice, Prescription, LabReport, PatientInsurance
Appointment(1)- (1) Invoice, Prescription, LabReport
Appointment(1)- (M) AppointmentReminder
Invoice(1) ---- (M) InvoiceItem, Payment
Invoice(1) ---- (1) InsuranceClaim
Prescription(1)-(M) PrescriptionItem -- Medicine
LabReport(1)---(M) LabReportItem -- LabTest
InsurancePanel(1)-(M) PatientInsurance, InsuranceClaim
Medicine(1) --- (M) StockMovement
PharmacyCategory(1)-(M) Medicine
LocumDoctor(1)-(M) LocumSession

```

### Tables Summary (33 migrations)

| #  | Table               | Key Columns                                        |
|----|---------------------|----------------------------------------------------|
| 1  | users               | name, email, role, branch_id, phone, profile_photo |
| 2  | branches            | name, code, address, phone, opening_time, closing_ |
| 3  | patients            | patient_id, name, ic_number, gender, dob, blood_ty |
| 4  | doctors             | user_id, branch_id, specialization, mmc_number, co |
| 5  | doctor_schedules    | doctor_id, day_of_week, start_time, end_time, slot |
| 6  | appointments        | patient_id, doctor_id, date, start_time, end_time, |
| 7  | services            | branch_id, name, price, category                   |
| 8  | invoices            | invoice_number, subtotal, tax, discount, total, st |
| 9  | invoice_items       | invoice_id, service_id, description, quantity, uni |
| 10 | payments            | invoice_id, amount, method, reference, payment_dat |
| 11 | insurance_panels    | company_name, type, credit_terms, consultation_lim |
| 12 | patient_insurances  | patient_id, panel_id, member_id, policy_number, re |
| 13 | insurance_claims    | claim_number, gl_number, claim_amount, approved_am |
| 14 | pharmacy_categories | branch_id, name, description                       |
| 15 | medicines           | name, sku, cost_price, selling_price, current_stoc |
| 16 | stock_movements     | medicine_id, type, quantity, stock_before, stock_a |
| 17 | prescriptions       | patient_id, doctor_id, appointment_id, status      |
| 18 | prescription_items  | medicine_id, dosage, frequency, duration, quantity |
| 19 | lab_tests           | name, category, normal_range, unit, price          |

|    |                       |                                                    |
|----|-----------------------|----------------------------------------------------|
| 20 | lab_reports           | report_number, patient_id, doctor_id, status       |
| 21 | lab_report_items      | lab_test_id, result, is_abnormal                   |
| 22 | appointment_reminders | appointment_id, channel, phone_number, message, st |
| 23 | locum_doctors         | name, ic_number, mmc_number, hourly_rate, session_ |
| 24 | locum_sessions        | locum_doctor_id, branch_id, session_date, total_pa |
| 25 | settings              | key, value                                         |
| 26 | audit_logs            | user_id, action, model_type, model_id, old/new_val |
| 27 | notifications         | user_id, type, title, message, icon, color, link,  |

## 7. Routes Reference

Total: ~150 routes under auth middleware.

### Core Routes

| Method   | URL                            | Description                  |
|----------|--------------------------------|------------------------------|
| GET      | /dashboard                     | Main dashboard with KPIs     |
| POST     | /branch/switch                 | Switch active branch         |
| CRUD     | /branches                      | Branch management            |
| CRUD     | /patients                      | Patient management           |
| POST     | /patients/{id}/portal-access   | Enable portal for patient    |
| CRUD     | /doctors                       | Doctor management            |
| GET/POST | /doctor-schedules/{doctor}     | Doctor schedule management   |
| CRUD     | /appointments                  | Appointment management       |
| PATCH    | /appointments/{id}/status      | Update appointment status    |
| CRUD     | /services                      | Service management           |
| CRUD     | /invoices                      | Invoice management           |
| GET      | /invoices/{id}/print           | Print invoice                |
| POST     | /invoices/{id}/payments        | Record payment               |
| CRUD     | /insurance-panels              | Insurance panel management   |
| CRUD     | /insurance-claims              | Insurance claim management   |
| PATCH    | /insurance-claims/{id}/status  | Update claim status          |
| CRUD     | /pharmacy-categories           | Pharmacy category management |
| CRUD     | /medicines                     | Medicine management          |
| POST     | /medicines/{id}/adjust-stock   | Adjust medicine stock        |
| CRUD     | /prescriptions                 | Prescription management      |
| PATCH    | /prescriptions/{id}/dispense   | Dispense prescription        |
| CRUD     | /lab-tests                     | Lab test management          |
| CRUD     | /lab-reports                   | Lab report management        |
| GET      | /lab-reports/{id}/print        | Print lab report             |
| CRUD     | /locum-doctors                 | Locum doctor management      |
| CRUD     | /locum-sessions                | Locum session management     |
| PATCH    | /locum-sessions/{id}/mark-paid | Mark session as paid         |
| GET/POST | /reminders                     | Reminder management          |
| POST     | /reminders/bulk                | Bulk create reminders        |

|           |                |                      |
|-----------|----------------|----------------------|
| GET/PATCH | /profile       | User profile         |
| DELETE    | /profile/photo | Remove profile photo |
| CRUD      | /users (admin) | User management      |
| GET/PUT   | /settings      | System settings      |

## Report Routes

| Method | URL                          | Description                 |
|--------|------------------------------|-----------------------------|
| GET    | /reports                     | Reports dashboard           |
| GET    | /reports/financial           | Financial report            |
| GET    | /reports/patients            | Patient report              |
| GET    | /reports/appointments        | Appointment report          |
| GET    | /reports/pharmacy            | Pharmacy report             |
| GET    | /reports/lab                 | Lab report analytics        |
| GET    | /reports/export/{type}       | CSV export (5 types)        |
| GET    | /audit-logs (admin)          | Audit log list              |
| GET    | /audit-logs/{id} (admin)     | Audit log detail            |
| GET    | /notifications               | Notification list           |
| POST   | /notifications/{id}/read     | Mark notification read      |
| POST   | /notifications/mark-all-read | Mark all notifications read |

## Patient Portal Routes

| Method | URL                      | Description                     |
|--------|--------------------------|---------------------------------|
| GET    | /portal/login            | Portal login page               |
| POST   | /portal/login            | Authenticate with IC + password |
| POST   | /portal/logout           | Logout                          |
| GET    | /portal/                 | Portal dashboard                |
| GET    | /portal/appointments     | View appointments               |
| GET    | /portal/invoices         | View invoices                   |
| GET    | /portal/invoices/{id}    | Invoice details                 |
| GET    | /portal/lab-reports      | View lab reports                |
| GET    | /portal/lab-reports/{id} | Lab report details              |
| GET    | /portal/prescriptions    | View prescriptions              |
| GET    | /portal/profile          | View profile                    |
| PATCH  | /portal/password         | Change password                 |

## 8. Patient Portal

A separate self-service portal for patients to view their medical records.

### Access

- URL: /portal/login
- Authentication: IC Number + Password (session-based, not Laravel auth)
- Enabling Access: Admin enables portal access from Patient management page

### Features

| Feature       | Description                                        |
|---------------|----------------------------------------------------|
| Dashboard     | Upcoming appointments, recent invoices, lab report |
| Appointments  | View all past and upcoming appointments            |
| Invoices      | View invoices with payment status and line items   |
| Lab Reports   | View completed lab results with abnormal flags     |
| Prescriptions | View medication details, dosage, instructions      |
| Profile       | View personal info, change password                |

The portal uses a separate Bootstrap navbar layout (not Star Admin sidebar) with shared Star Admin CSS/JS for consistent styling. It is fully mobile-responsive.



## 9. Audit & Security

### 9.1 Audit Logs

Every significant action is automatically tracked via the Auditable trait.

- Tracked Actions: Create, Update, Delete
- Tracked Models (14): Patient, Appointment, Invoice, Doctor, Branch, Service, Medicine, Prescription, LabReport, InsurancePanel, InsuranceClaim, LocumDoctor, LocumSession, User
- Logged Data: User, Branch, Action, Model, Description, Old Values (JSON), New Values (JSON), IP Address, User Agent, Timestamp
- Sensitive Fields Excluded: password, remember\_token, portal\_token
- Access: Admin only - filterable by user, action, module, date range

### 9.2 In-App Notifications

Automatic alerts for important events via the NotifiesUsers trait.

| Event                  | Notified Roles         | Type        |
|------------------------|------------------------|-------------|
| New appointment booked | Doctor, Admin, Recept. | appointment |
| Appointment cancelled  | Admin, Receptionist    | appointment |
| Invoice paid           | Admin                  | invoice     |
| Claim status change    | Admin                  | insurance   |
| Prescription dispensed | Admin, Doctor          | pharmacy    |
| Lab report completed   | Doctor                 | lab         |
| Low stock alert        | Admin, Staff           | pharmacy    |

- Bell Icon: Navbar dropdown with 5 most recent + unread count badge
- Full Page: Filterable by type and read/unread status
- Actions: Mark as read, mark all read, delete, click-through to related record

### 9.3 Security Features

- RBAC: Role-based middleware on sensitive routes
- CSRF Protection: All POST/PUT/DELETE forms
- Password Hashing: bcrypt via Laravel
- Session-based Auth: Staff (Laravel session) + Portal (custom session)
- Inactive User Block: Deactivated users cannot log in
- Admin-Only Operations: User management, audit logs

## 10. Reports & Analytics

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5 comprehensive report modules with filtering and CSV export.

### 10.1 Financial Report

- KPIs: Total Revenue, Outstanding Amount, Invoice Count, Paid Count
  - Charts: Revenue by Branch, Revenue by Doctor, Revenue by Service (Top 15)
  - Breakdown: Payment Method distribution
  - Trend: Daily Revenue line chart
  - Filters: Branch, Date Range | Export: CSV
- 

### 10.2 Patient Report

- KPIs: Total Patients, New This Month, Active Patients, Insured Count
  - Charts: Gender Distribution, Blood Type Distribution, Monthly New Patients (6 months)
  - Table: Top 10 Patients by Visits
  - Filters: Branch | Export: CSV
- 

### 10.3 Appointment Report

- KPIs: Total Appointments, Completed, Cancelled, Completion Rate
  - Charts: Status Breakdown, Peak Hours Analysis, Day of Week, Daily Trend
  - Table: Appointments by Doctor with completion stats
  - Filters: Branch, Date Range | Export: CSV
- 

### 10.4 Pharmacy Report

- KPIs: Total Medicines, Low Stock, Expiring Soon, Expired, Stock Value
  - Tables: Low Stock Alert List, Expiring Soon List, Top 10 Dispensed (30 days), Stock by Category
  - Filters: Branch | Export: CSV
- 

### 10.5 Lab Report Analytics

- KPIs: Total Reports, Completed, Pending
- Charts: Status Breakdown, Daily Volume Trend
- Table: Top 10 Tests Ordered with abnormal count
- Filters: Branch, Date Range | Export: CSV

## 11. Default Test Data

The database seeder creates a complete test environment for development and demo.

### Branches

| Branch                   | Code  | Hours       |
|--------------------------|-------|-------------|
| Klinik Utama - Shah Alam | KU-SA | 08:00-22:00 |
| Klinik Utama - Subang    | KU-SB | 08:00-21:00 |

### Users

| Name               | Role   | Email                   |
|--------------------|--------|-------------------------|
| Admin              | Admin  | admin@clinic.com        |
| Dr. Ahmad Razak    | Doctor | dr.ahmad@clinic.com     |
| Dr. Siti Nurhaliza | Doctor | dr.siti@clinic.com      |
| Dr. Lee Wei Ming   | Doctor | dr.lee@clinic.com       |
| Nurse Aminah       | Staff  | nurse.aminah@clinic.com |

### Patients (5)

Muhammad Ali bin Hassan, Fatimah binti Abdullah, Tan Mei Ling, Rajesh a/l Kumar, Nurul Aina binti Ismail

### Services (10 per branch)

General Consultation (RM50), Specialist Consultation (RM120), Blood Test (RM80), X-Ray (RM150), ECG (RM100), Wound Dressing (RM40), Injection (RM30), Nebulizer (RM35), MC Letter (RM10), Referral Letter (RM15)

### Insurance Panels (4 per branch)

AIA Bhd, PETRONAS Corporate Panel, Great Eastern Life, SOCSO (PERKESO)

### Pharmacy

4 categories x 8 medicines per category = 32 medicines per branch. Categories: Antibiotics, Painkillers, Vitamins & Supplements, Cough & Cold

### Lab Tests (10 per branch)

FBC, Fasting Blood Sugar, HbA1c, Lipid Profile, Liver Function Test, Renal Profile, Urine FEME, Uric Acid, Thyroid Function Test, Chest X-Ray

## **Locum Doctors (2)**

Dr. Zainab binti Yusof, Dr. Krishnan a/l Raju

## Appendix: Auto-Generated ID Formats

| Entity     | Format                      | Example          |
|------------|-----------------------------|------------------|
| Patient ID | {branch_code}-P{5-digit}    | KU-SA-P00001     |
| Invoice No | INV-{branch_code}-{6-digit} | INV-KU-SA-000001 |
| Claim No   | CLM-{branch_code}-{5-digit} | CLM-KU-SA-00001  |
| Lab Report | LAB-{branch_code}-{5-digit} | LAB-KU-SA-00001  |

*End of Documentation*